Almost-all non-halting for Busy-Beaver renewal maps

A transfer-operator attack on the BB(6) cryptid cohort, in the style of Tao's Collatz theorem

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Problem. The determination of BB(6) is obstructed by a cohort of ∼122 "cryptid" holdout machines—Antihydra the flagship—whose halting is equivalent to a Collatz-type first-passage event for an explicit integer map [2, 3]. Deciding any single such orbit is Collatz-hard. We pursue the statistical question instead, exactly as Tao did for Collatz [1]: among all N^2 integer seeds, what fraction halt?

Approach. Each machine's hurler/catcher macro-dynamics reduces to a piecewise-affine *renewal map* $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ whose first return to a section is a countable full-branch *Gauss-type expanding Markov map* G (branch endpoints $\rho_{k+1} = (3\rho_k + 1)^{-1}$, accumulation $\rho^* = (\sqrt{13} - 1)/6$). Halting $\iff$ the orbit meets a fatal set $\mathcal{D}$. To this object we apply the modern transfer-operator stack: Lasota–Yorke/Rychlik for the invariant measure [4, 5], a Dolgopyat-type uniform spectral gap for the twisted operator [6, 7], and the Nagaev–Guivarc'h local limit theorem [8, 9], with Keller–Liverani stability for the model-to-exact and countable-branch truncations [10].

Results.

- *Measure form (unconditional).* Lebesgue-almost every real seed is non-halting; every orbit has $a + b \to \infty$. Proved via Rychlik's theorem [5], all hypotheses (full branches, eventual expansion $\inf |(G^2)'| = 3$, summable variation) verified from the exact branch arithmetic.
- *Density form (reduced).* $\#\{\text{halts} \leq N\} = O(N \log N) = o(N^2)$, conditional on the spectral gap of Naud's Theorem 2.3 [7]—whose non-local-integrability hypothesis we *verify in closed form* (the branch roofs $L_k(\xi) = (r_k - p_k) + (s_k - q_k)\xi$ are pairwise non-proportional, slopes $-\rho_k$ distinct)—and the Nagaev–Guivarc'h LLT ($\sigma^2 \approx 0.265 > 0$, non-lattice, both verified). The countable-branch gap reduces to the finite-branch one via an *exact* geometric tail (rate $\lambda^{-1} = 1 - \rho^* = 3\rho^{*2}$); the gap holds uniformly in scale because the Dolgopyat hypotheses are scale-independent algebra of the branch matrices.
- *Remaining core.* The density form rests, finally, on one arithmetic exponential-sum estimate—the direct analogue of Tao's characteristic-function bound on $\mathbb{Z}/3^n\mathbb{Z}$ [1]. A direct probe shows the relevant cross-cylinder cancellation is *friendly* (empirical sum at the sampling-noise floor, resonances only at low 3-adic frequencies), reframing the estimate from adversarial to hard-but-standard.
- *Mechanization.* A fail-closed *grammar extractor* recovers each machine's symbolic catch-grammar and its forcing parity invariant with a Lean-checkable completeness obligation (validated on the renewal map: it re-derives the proven grammar and the non-intermittency that certifies the quadratic fuel bound). This scales the 18 existing kernel-checked conditional closures (Safe → ¬Halt, axioms {propext, Quot.sound}) toward the full cohort.

Honest status & what we seek. The measure form is complete and unconditional; the density form is a rigorous reduction to cited theorems with every dynamical hypothesis discharged, open only at the one arithmetic estimate above; single-machine halting remains Collatz-hard and untouched. This is the three-tier posture of Tao's theorem relative to Collatz. We seek feedback on (i) closing the cross-cylinder exponential-sum estimate, and (ii) the grammar-certified decider as a route to cohort-scale conditional closure.

References

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